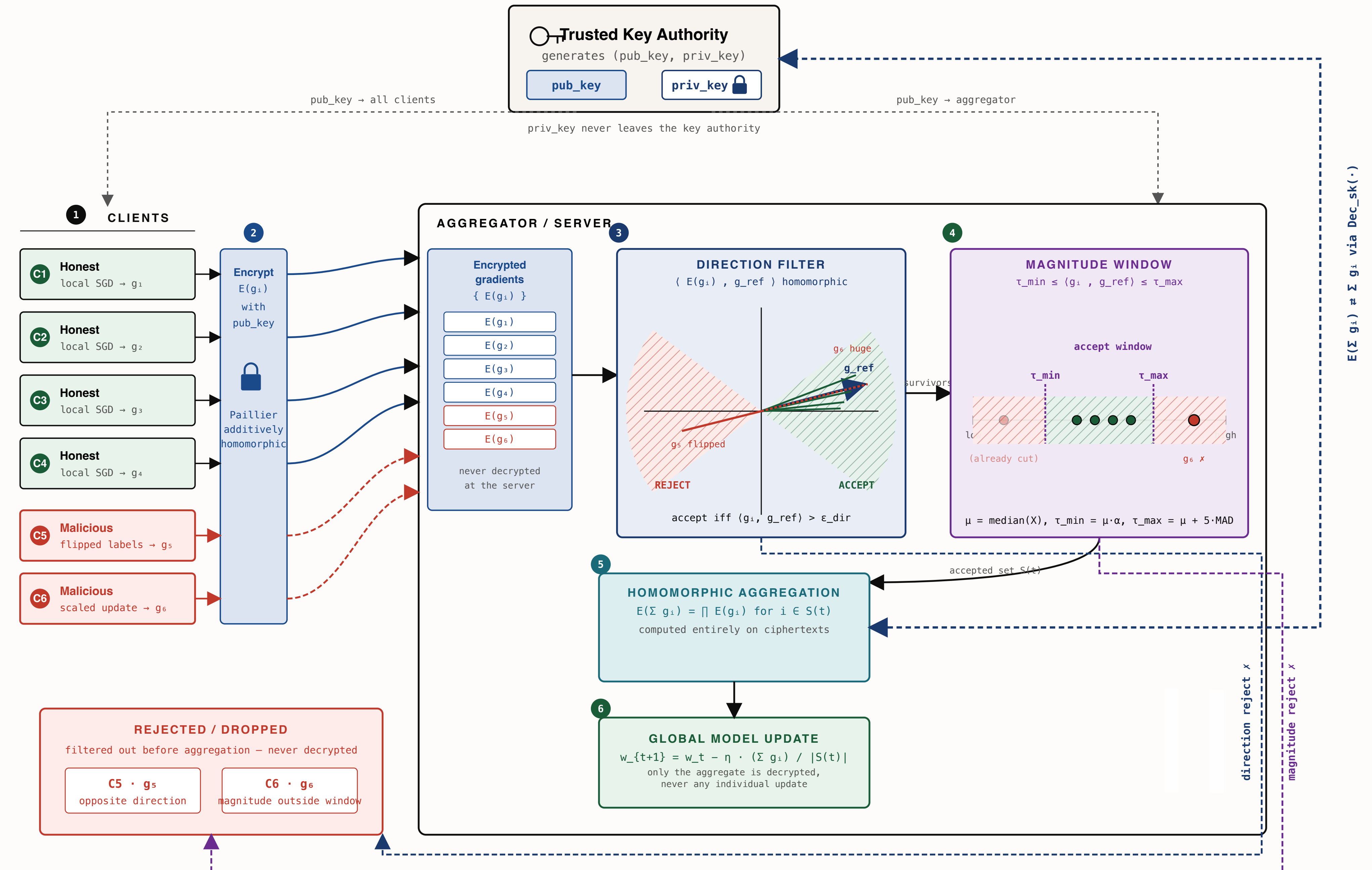




FLOW ① local training → ② Paillier encryption → ③ direction filter → ④ magnitude window → ⑤ homomorphic aggregation → ⑥ decrypted global update

Privacy property: every per-client gradient remains encrypted end-to-end. Only one scalar (the audit score) and the aggregate are ever decrypted, both via priv_key held only by the Trusted Key Authority.